



Igniters Innovation Challenge 2026 | Engineering Track

An exclusive Engineering Track

Hosted by: Suzuki R&D Center

TEAM WOOKIES

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DON'T BUILD HYDROGEN CARS. CONVERT THE 7.5 MILLION INDIA ALREADY OWNS.

H₂-SHIFT | A bi-fuel CNG → Hydrogen retrofit platform — zero range anxiety, zero wait for infrastructure.

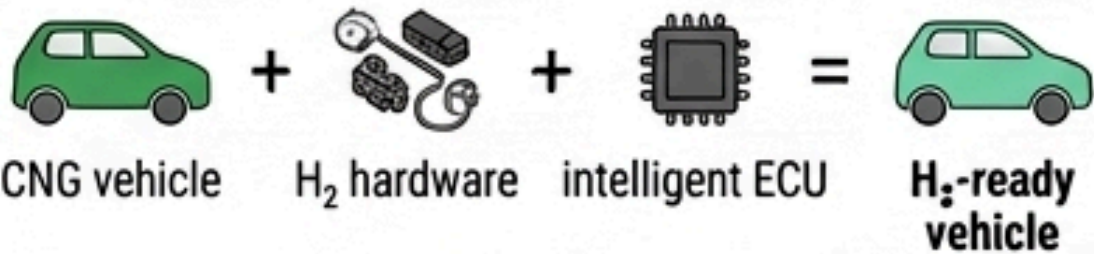
1. Market insight → 2. Retrofit Ladder → 3. Vehicle architecture & safety → 4. Why this beats replacement

WHY CNG IS INDIA'S HIDDEN HYDROGEN ASSET

7.5 Mn
India's CNG vehicle fleet
3x growth in 8 years

~65%
Passenger vehicles
Natural retrofit pool

~71%
Maruti Suzuki share of India's CNG passenger market
6.2 lakh CNG cars sold FY25



Why CNG → H₂ is the shortest technical jump

Spark-ignition engine ✓

Gaseous injection ✓

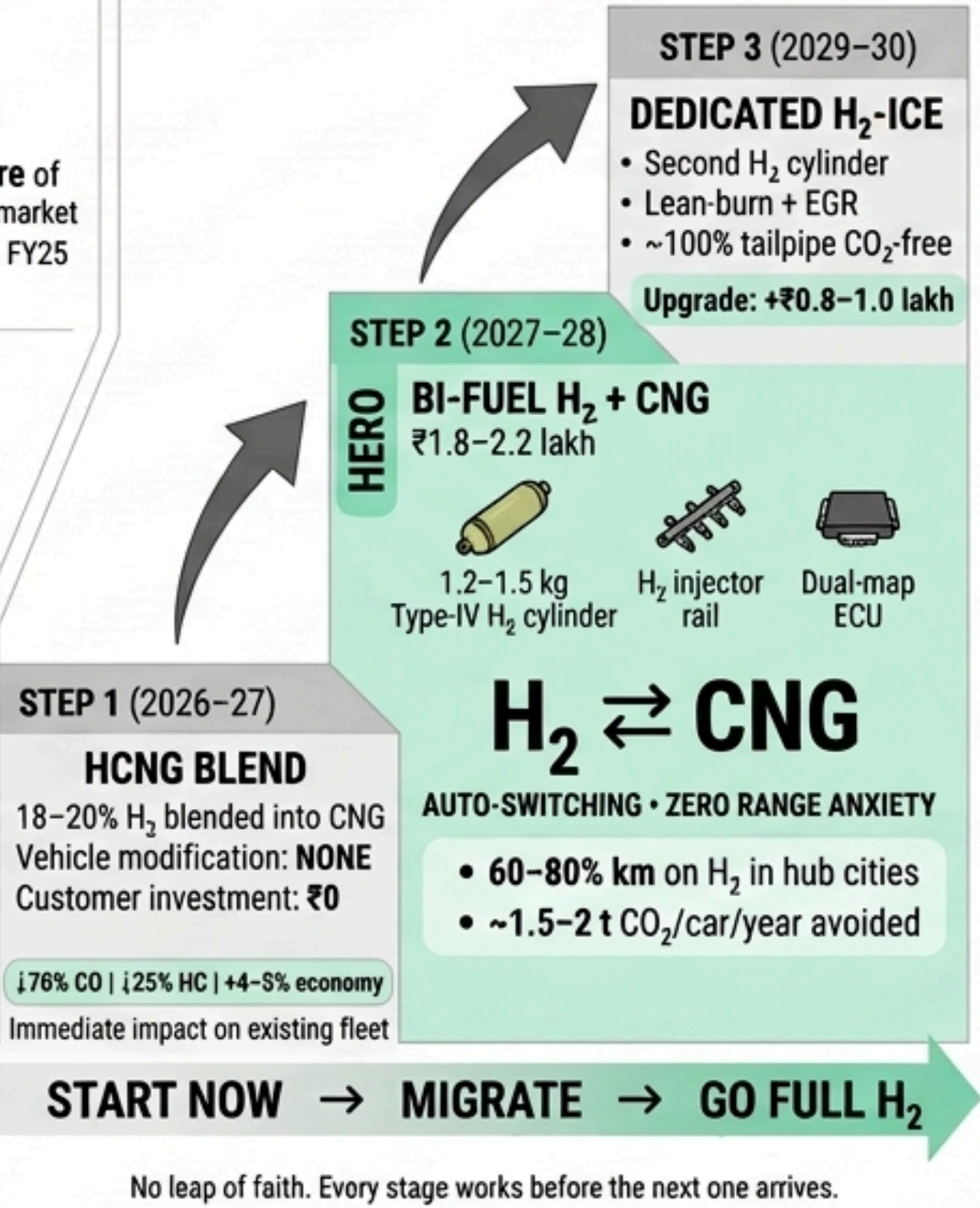
Existing cylinder packaging ✓

4,000+ retrofit centres ✓

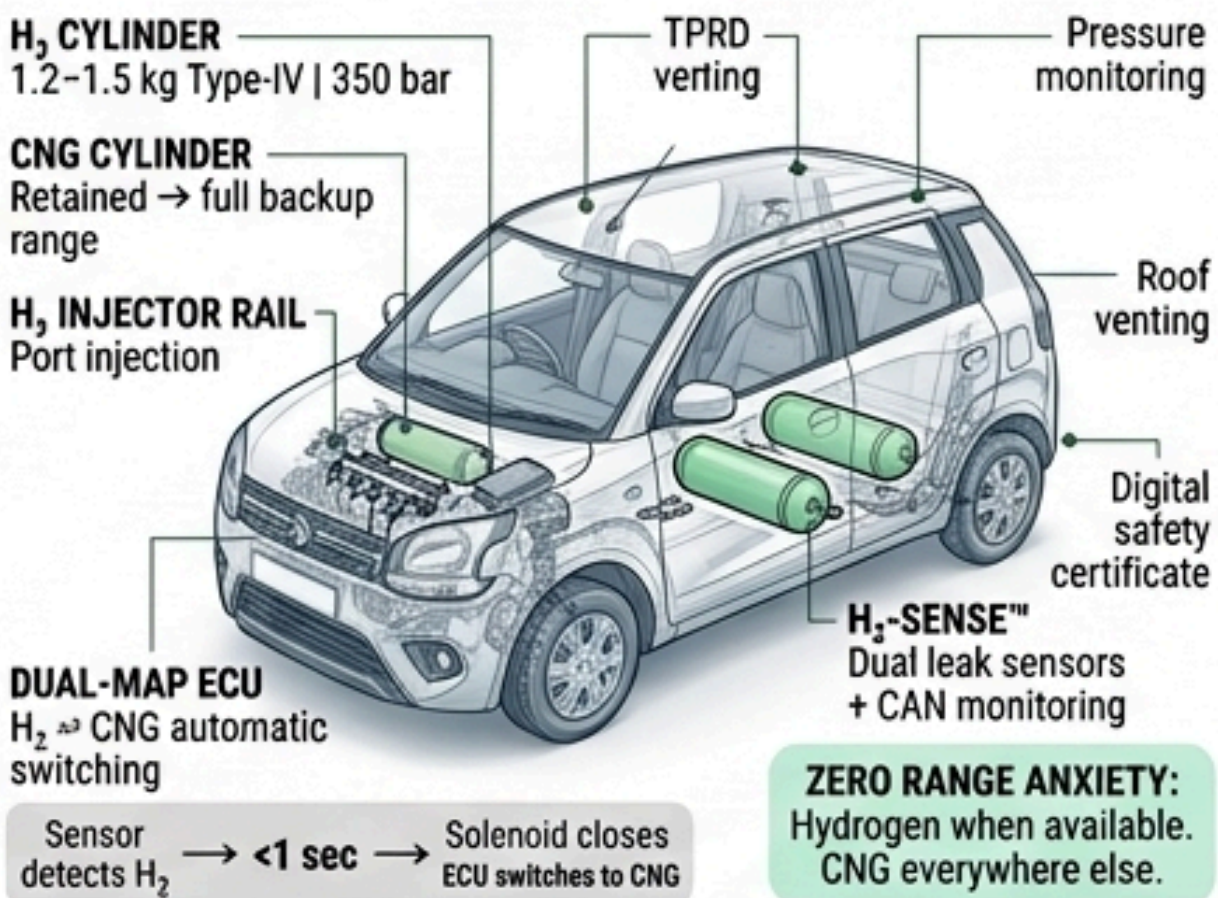
Full testing cylinder safety certifications →

"A petrol → H₂ conversion rebuilds the car.
A CNG → H₂ conversion upgrades it."

THE RETROFIT LADDER™



WHAT ACTUALLY CHANGES IN THE CAR?



RETROFIT vs REPLACEMENT		
METRIC	H ₂ -SHIFT	NEW BEV
UPFRONT COST	₹1.8-2.2L	₹9-14L
RANGE / REFUEL	400+ km combined 3-5 min	220-350 km ~45 min fast charge
DAY-1 INFRASTRUCTURE	7,400+ CNG stations	Charging network required
EMBODIED CO ₂	Extends existing car life	~5-8 t CO ₂ before km 1
CUSTOMER	Existing vehicle owner	New-car buyer
BEVs replace cars. H ₂ -SHIFT decarbonises the installed base.		

FLEET-ECONOMICAL TODAY, HYDROGEN-CHEAP TOMORROW – AND IT SPINS THE MISSION'S FLYWHEEL.

Bi-fuel architecture removes today's infrastructure constraint while creating the demand that makes tomorrow's H₂ economics possible.

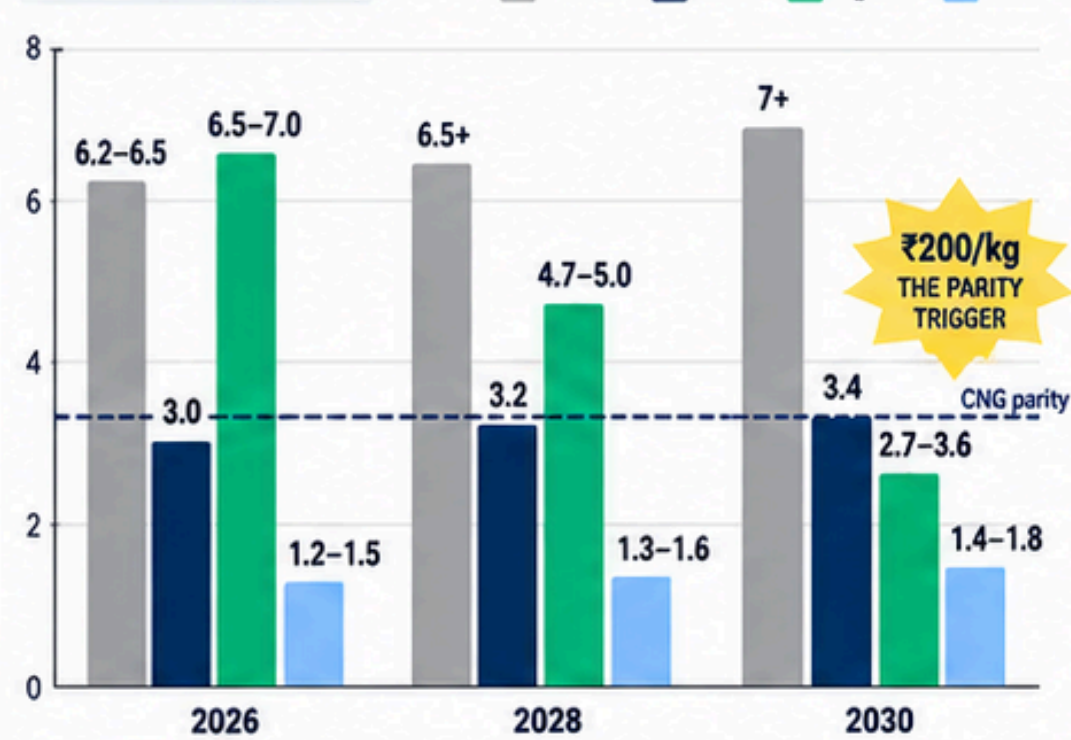
H₂-SHIFT

Convert. Don't Replace.

1. THE ₹200 PARITY TRIGGER

Hydrogen isn't cheap today — the architecture doesn't require it to be.

RUNNING COST (₹/km)



KEY TAKEAWAYS

- H₂ cost declines, crosses CNG by 2030
- Bi-fuel lets you capture value at every price point
- Lower running cost than petrol from Day 1 for fleet users

GREEN HYDROGEN PRICE TRAJECTORY (₹/kg)



At ₹200/kg delivered H₂, running cost ≈ CNG.

FLEET TAXI | 150 KM/DAY

~₹1.5L annual saving vs petrol

~18-month kit payback

Buy before parity. Auto-switch fuel priority when parity arrives.

2. SAFETY BY DESIGN

Different fuel, different risks — engineered mitigation for hydrogen.

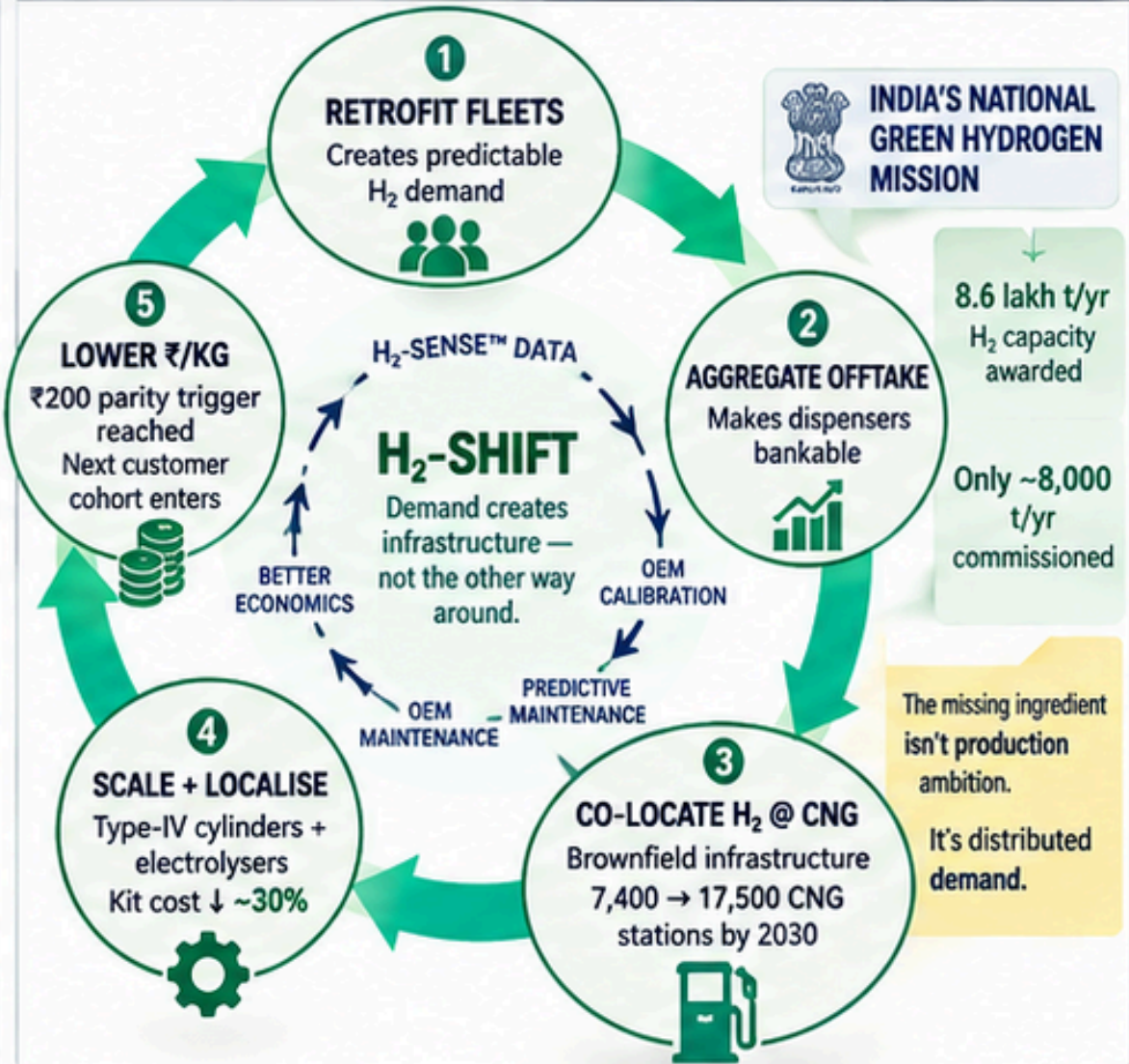
PARAMETER	PETROL	LPG	CNG	H ₂ + H ₂ -SENSE™
LEAK BEHAVIOUR	Pools	Pools	Rises	Disperses ~4x faster
IGNITION RISK	Low energy	Low	Moderate	Very low ignition energy — actively managed
STORAGE	Conventional tank	Steel cylinder	Type-I/II	Type-IV composite
PROTECTION	Basic	Basic	PRD	• Dual sensors • < 1 sec shut-off • TPRD • Roof venting • ≥ 2.25x burst test
REGULATION	CMVR	CMVR	CMVR + PESO	AIS-157 + PESO + ISO / SAE

⚠ Hydrogen has different risks, not higher risks — with the right design.

We inherit 20 years of India's CNG homologation & retrofit safety learning.

3. THE H₂-SHIFT FLYWHEEL™

A virtuous cycle from demand to a self-sustaining ecosystem.



4. OUR ROADMAP TO 2030

From pilots to a self-sustaining ecosystem

2026-27 | IGNITE



- HCNG + certification pilots
- Delhi-NCR + Surat
- 100 stations
- 500 fleet cars
- ARAI / iCAT certification

2027-28 | CERTIFY



- Commercial bi-fuel rollout
- AIS / PESO approval
- NBFC financing
- Fleet aggregator partnerships
- 25,000 kits | 5 cities

2029-30 | SCALE



- ₹200 parity trigger
- Retail launch
- 8 H₂ Halo cities
- Localised kit manufacturing
- **1,00,000+ CARS**

THE IMPACT BY 2030



1,00,000+ cars converted



~0.9 Mt total CO₂ avoided



~15,000 t/yr firm green-H₂ demand



₹2,000+ Cr retrofit industry unlocked



10,000+ certified green jobs

Same Cars
Bigger Impact



H₂-SHIFT doesn't wait for India's hydrogen ecosystem to mature — it creates the demand that makes the ecosystem mature.

ASSUMPTIONS, BENCHMARKS & STANDARDS

Every critical technical and economic assumption is anchored to an **observable** benchmark or certification pathway.

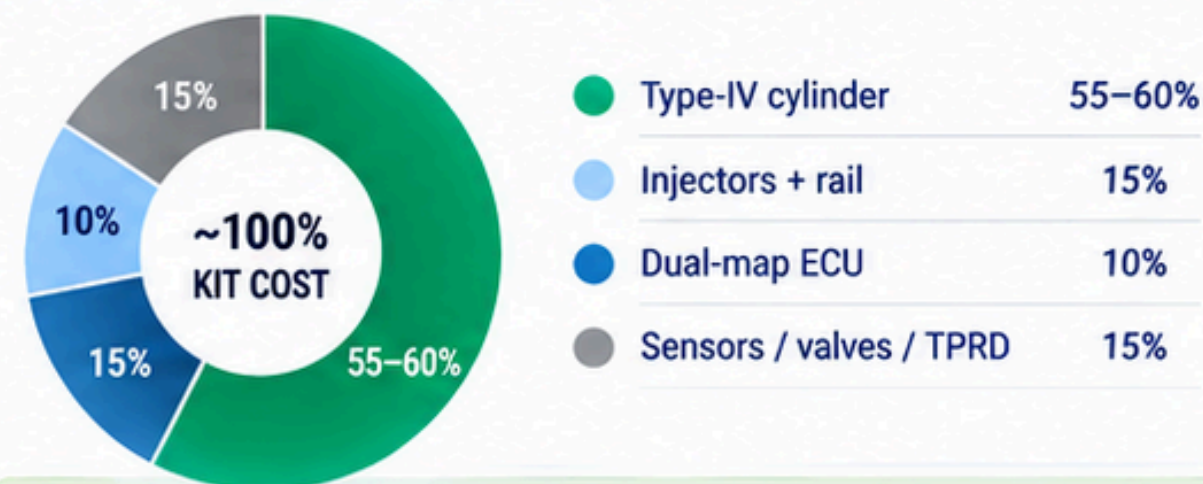
Same Cars
Bigger Possibilities

H₂-SHIFT
Convert. Don't Replace.

1 KEY MODEL ASSUMPTIONS Conservative, credible, and decision-relevant

PARAMETER	ASSUMPTION	BASIS
 H ₂ -ICE mileage	55–60 km/kg	Energy density + retrofit studies
 Green H ₂ price (₹/kg)	₹397 → ₹260–310 → ₹160–200	2026 → 2268 → 2030 (cost decline trajectory)
 Bi-fuel kit price	₹1.8 – 2.2 L	Type-IV cylinder = largest BOM component
 CNG price & mileage	₹78/kg 26 km/kg	Current market levels
 Fleet duty cycle	150 km/day 330 days	Taxi / fleet benchmark
 Carbon credits	\$10–15 / t CO ₂	Voluntary market range

KIT BILL OF MATERIAL (Indicative)



 **#1 COST-DOWN LEVER**
→ **LOCALISE TYPE-IV CYLINDERS**

Drives 20–30% lower kit cost at scale

2 GLOBAL PROOF POINTS Real projects. Real learning. Real confidence.



TOYOTA
H₂-ICE GR Corolla + Hilux prototypes

- **Proof point:** Hydrogen combustion
- durability





SUZUKI
Hydrogen Burgman Street prototype | 70 MPa

- **Proof point:** Suzuki already has H₂-ICE DNA
- **Strategic fit:** Leverages Suzuki's global hydrogen development





INDIA — EARLY DEPLOYMENTS SHOW TRACTION

- 70 pilot vehicles + 16 H₂ stations
- ₹410.77 Cr mobility pilots
- IOCL 50-bus HCNG trial (with patented compact reformer)





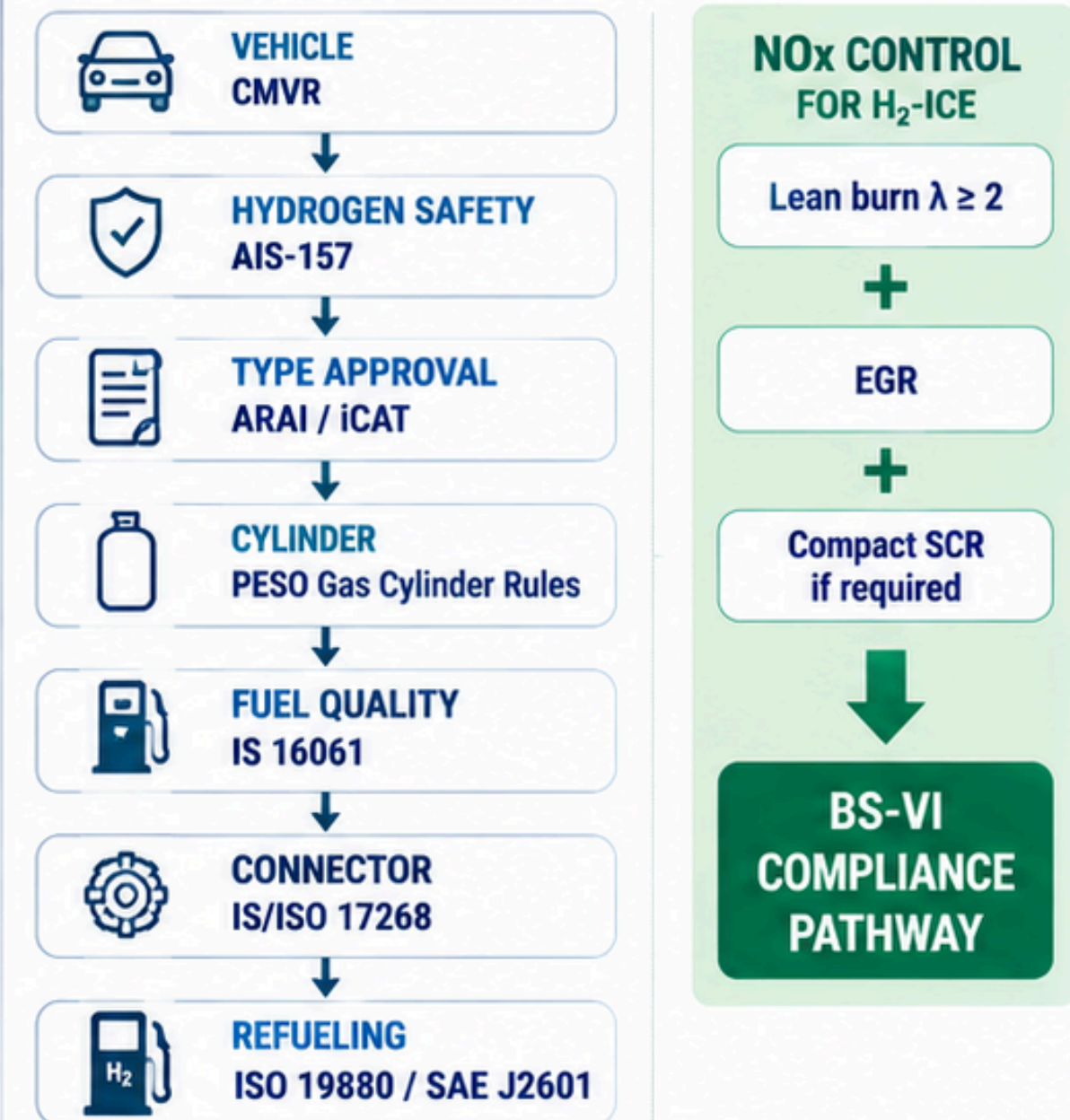
ACADEMIA — TECHNOLOGY VALIDATION

- Small-car retrofit studies demonstrate:
- Comparable range / power
- PFI = simplest architecture
- Injection phasing manages backfire

REAL-WORLD STUDIES DER-RISK THE TECHNOLOGY

 **KEY TAKEAWAY**
H₂-SHIFT combines proven components into a new Indian retrofit architecture — rather than betting on unproven physics.

3 REGULATORY PATHWAY Clear standards. Proven pathway. Faster to market.



 **Regulatory strategy:** mirror India's proven CNG retrofit homologation pathway rather than create a new route from scratch.

SOURCES (1/2)

1. CRISIL – India CNG outlook
2. Govt. of India – H₂ pilot projects
3. Maruti Suzuki FY25 investor update
4. IOCL – HCNG reformer patent
5. Toyota – H₂-ICE prototypes
6. Suzuki – Hydrogen Burgman prototype
7. Academic studies on H₂ retrofits
8. Global H₂ cost outlook (IEA / BNEF)
9. AIS-157 draft standard
10. PESO Gas Cylinder Rules
11. IS 16061 – H₂ fuel quality
12. ISO 19880 / SAE J2601 – Refueling
13. Carbon credit price benchmarks

Same Standards
Greater Impact



A SAFER, CLEANER, STRONGER INDIA — TOGETHER